

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 63



SCIENCE
&
TECHNOLOGY-11



WHY IS SEMAGLUTIDE MOLECULE IN THE NEWS RECENTLY?

- Semaglutide molecule, used to treat type 2 diabetes and/or weight loss went off-patent on 20 March 2026.
- Now, generic versions are flooding the market and are selling at 50 to 80% lower than the price for either of the two patented brands Ozempic and Wegovy.
- The monthly cost of these generics is said to drop from Rs. 12,000 to Rs. 3,000 or below making it accessible not just to middle-class, middle-aged obese diabetics but also to younger people who want to look good.



- Semaglutide belongs to a class of medicines called GLP-1 receptor agonists. These drugs engineer the brain to curb hunger, stimulates insulin release when blood sugar levels are high, and slows down digestion.
- There is no denying its effectiveness in reducing weight loss over a few months, and in curbing sugar levels. It is recommended for those with a body mass index of above 30 kg per sq m, or 27 kg per sq m with a medical condition. Such individuals need to shed weight quickly to reduce the risk of a stroke, or heart attack.
- The drug is essentially administered as a weekly injectible (2.4 mg dosage). It is not recommended for those with a family history of thyroid cancer.
- In rare cases, it could lead to blindness. It is not to be used by those who are not medically obese, and that includes diabetics.

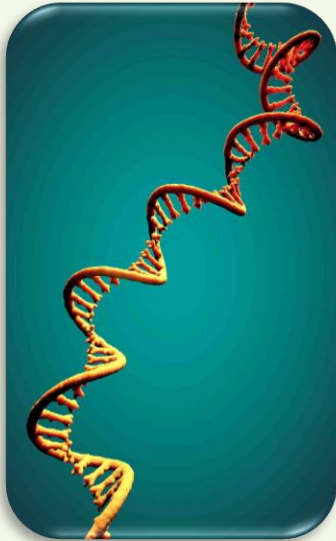
As the patent on Wegovy and Ozempic expires, GLP-1 drugs may flood Indian market

• The price war

Natco Pharma	2 mg/1.5ml	4 mg/3 ml	8 mg/3 ml		
Cost of vial	Rs 1,290	Rs 1,290	Rs 1,750		
Cost of pen device	Rs 4,000	Rs 4,200	Rs 4,500		
Eris Lifesciences	2mg/1.5 ml	4 mg/3 ml	8 mg/3 ml		
Cost of vial	Rs 1,290	Rs 1,290	N.A.		
Cost of pen device	Rs 4,000	Rs 4,200	Rs 4,500		
Novo Nordisk	0.25mg/1.5ml	0.5mg/1.5ml	1mg/3ml	1.7mg/3ml	2.4mg/3ml
Cost of Wegovy pen device	Rs 10,850	Rs 13,850	Rs 13,850	Rs 16,400	Rs 16,400
Emcure Pharma*	0.25mg/1.5ml	0.5mg/1.5ml	1mg/3ml	1.7mg/3ml	2.4mg/3ml
Cost of Poviztra pen device	Rs 8,790	Rs 11,200	Rs 11,200	Rs 13,000	Rs 15,000

PRICES FOR A MONTH OF TREATMENT, *LAUNCHED IN DEC 2025 IN TIE-UP WITH NOVO, SOURCE: PHARMARACK, COMPANIES

The discovery of exRNA changed modern medicine, for instance, just by testing a patient's blood or other body fluids, doctors can identify specific RNA patterns linked to cancer or heart disease.



- ex-RNA is RNA that exists outside cells in body fluids, such as blood, saliva, urine, and cerebrospinal fluid.
- Scientists recently reported that extracellular RNA (exRNA) from bacteria can persist in disinfected drinking water. They also found that by studying the exRNA, they could figure out what the bacteria were doing just before they were damaged or killed, releasing the exRNA. This way, the scientists could figure out which survival strategies worked for the bacteria, which can be used to make better disinfectants.
- For decades, scientists believed RNA only functioned inside the cell and assumed that if RNA leaked out, enzymes in the blood would destroy it. However, researchers have found that cells, in fact, intentionally export RNA.
- Now, it is found that a cell releases RNA to deliver instruction to another cell elsewhere in the body, changing how it behaves or which genes it activates. This process helps coordinate responses in the immune system, tissue repair, and development. However, cancer cells can also release exRNA to promote tumor growth.



DOMESTIC LPG CONNECTIONS MORE THAN DOUBLED IN 10 YEARS, THANKS TO UJJWALA!

- Domestic LPG connections grew from 150 million in 2015 to 332 million by 2025.
- India imports around 60% of its LPG and around 50-60% of its natural gas. The combined import bill hit \$26.4 billion in FY24-25 - a 50% jump in six years.
- As per IEEFA, electric cooking is 37% cheaper than non-subsidised LPG and 14% cheaper than piped natural gas for a family of four in Delhi – even without any electricity subsidy.
- Induction cooktops transfer about 85% of energy to the vessel, while an LPG burner manages roughly 40%.

Electric pressure cookers use less energy than any other device assessed. There is 5% electric cooking share in 2021, as per reports available.



- The most popular electrified cooking equipment is the induction stove. It consists of a glass surface covering the heating component.
- When an alternating electric current passes through a copper coil beneath the glass, it creates a fluctuating magnetic field. When you place a magnetic pot on top of the surface, the field induces electric currents inside the metal. These currents meet resistance, generating heat directly in the vessel instead of the stove top.
- A gas flame is more inefficient because it loses nearly 60% of its heat to the surrounding air, but an induction stove can be about 85-90% efficient because it uses magnetism to heat the vessel directly without warming the air.
- To switch to cooking with an induction stove alone, users have to pay for the cooktop, which usually ranges from Rs. 2,000 to Rs. 4,000. They will also need to pay for induction compatible cookware such as stainless steel or cast-iron pans with flat bottoms.

For commercial establishments like hotels, there is a need to have expensive high-tension connections. At present, several restaurants have low tension current transformer connections. The capital cost for electrical equipment is two to three times more compared with those used for LPG cylinders.

OPEN AUTOMATED DEMAND RESPONSE (Open ADR) IS THE ADVANCED SYSTEM TO TAKE CARE OF PEAK POWER DEMAND



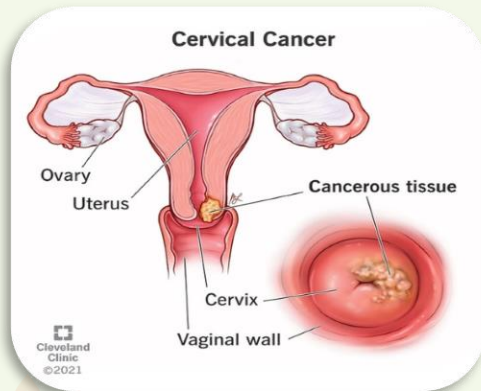
- Open ADR automates participation of smart thermostats, EV chargers, water heaters and cooktops in demand response. These devices can adjust their consumption automatically once we switch over to open ADR.
- First open ADR system was brought as a pilot project in Delhi by Tata Power Distribution Company.
- What we need is open ADR compliant servers, smart meter embedded receivers, and aggregator platforms that can orchestrate distributed loads into virtual power plants.

WHAT ARE THE OTHER SOLUTIONS AVAILABLE?



- A rooftop solar panel paired with a battery can turn a household into a prosumer.
- Peer-to-Peer (P2P) energy trading allows households to sell excess electricity directly to neighbors through digital platforms, bypassing the traditional discom route. India ran South Asia's first blockchain based P2P solar trading pilot in Lucknow, led by India Smart Grid Forum and Australia's Powerledger.



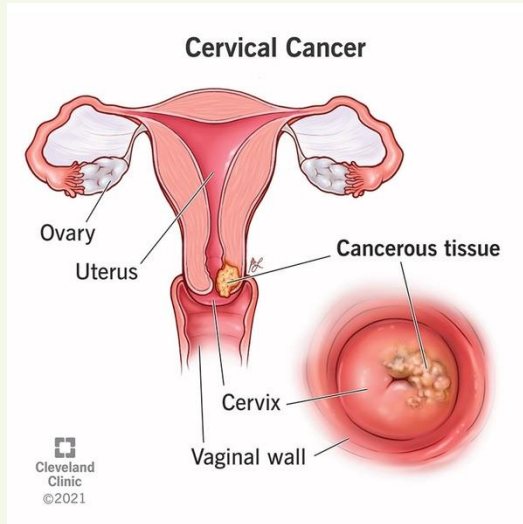


It is mandatory to seek consent from parents or guardians before giving the shot. Hence, this requires an opt-in consent rather than opt-out method followed for all other childhood vaccines.

PRIME MINISTER ROLLED OUT HPV VACCINE CAMPAIGN FROM RAJASTHAN ON FEBRUARY 28

- The special nation-wide vaccination campaign against Human Papilloma Virus (HPV), which causes cervical cancer, will be flagged off by the Prime Minister on February 28 from Rajasthan's Ajmer.
- All 14-year-old girls and those who turn 15 during the following 90 days will be eligible to receive the shot.
- Cervical cancer continues to be the second most common form of cancer in Indian women, affecting 1.25 lakh and killing 75,000 each year.
- The vaccine will be available in all days for the first 90 days of the campaign through nearly 30,000 immunization centers across the country, located in government health centers and hospitals.
- There was no additional challenge with storing the HPV vaccines, as they have to be maintained at a 2-8 degrees Celsius temperature range, which is the common setting for most other vaccines under the government's universal immunization program.

WHAT DO YOU UNDERSTAND BY CERVICAL CANCER? WHY IS IT CONSIDERED A MAJOR PROBLEM FOR INDIAN WOMEN?



- Cancer of the Cervix (literally, the neck of the womb) is unique among cancers because almost 99% of the cases are linked to infection with the Human Papillomavirus (HPV), a common virus transmitted through sexual contact. HPV do not contain the lipid envelop.
- HPV is the name for common group of viruses, there are more than 100 types of HPV. Many women will be infected with HPV during the course of their lifetime, with no ill-effect.
- While most HPV infections resolve spontaneously and the women remain symptom-free, persistent infection can lead to cervical cancer.
- It is the second leading cause of cancer-related deaths among women in India. Annually, more than 77,000 deaths are happening.
- It is the second most frequent cancer among Indian women between 15 and 44 years, after breast cancer.

WHAT IS WHO STRATEGY?

Close to 90% of global cervical cancer deaths occur in Low- and Middle-Income Countries (LMICs).



- WHO's strategy outlines a 90-70-90 triple pillar intervention to be implemented by 2030 with an additional focus on high-quality health care and equitable health care services.
 - ✓ 90% of girls must be fully vaccinated with the HPV vaccine by the age of 15.
 - ✓ 70% of women must be screened using a high-performance screening test by the age of 35 and again by 45.
 - ✓ 90% of women detected with cervical pre-cancer and cancer lesions must receive treatment and care.
- The target for elimination of cervical cancer as a public health problem is to reduce cervical cancer incidence to less than four cases per one lakh woman years. These resolutions shaped the global elimination strategy and set clear targets to be achieved by 2030.

The screening for Cervical Cancer in India is just under 2%.

The HPV Vaccine was introduced in India in 2008. It was recommended for inclusion in the Universal Immunization Programme in 2023.



1. The program will use a single dose of Gardasil for protection from HPV type 16 and 18 which cause cervical cancer and types 6 and 11.
2. India has secured HPV vaccine supplies through a transparent, globally supported procurement mechanism under India's partnership with Gavi, the Vaccine Alliance.
3. HPV vaccines are among the most extensively studied vaccines worldwide, with evidence demonstrating 93-100% effectiveness in preventing cervical cancer caused by vaccine-covered HPV types.
4. HPV virus is a double-stranded DNA virus. The vaccine is non-live, does not cause HPV infection. This vaccine can be given to boys also. HPV types 16 and 18 are responsible for nearly 70% of cases worldwide.

WE NEED MULTI-PRONGED STRATEGY FOR DESIRABLE OUTCOMES

ning Space



- HPV vaccine can be given to women up to age 45 years, but its effectiveness will be less. It is highly effective when given before exposure to the virus.
- HPV vaccine does not replace other medical examinations. Vaccination and screening must go hand in hand.
- No vaccine currently protects against all cancer-causing HPV types. This means there remains a small but significant risk of cervical cell changes even after vaccination.
- Once the woman crossed 30, It is recommended to undergo both the cervical screening tests and HPV-DNA testing.
- A pap smear or cervical cytology test detects early abnormal changes in cervical cells long before symptoms appear. These changes can be treated effectively if identified on time, preventing progression to cancer.



IIT MADRAS RELEASED A GENOME DATABASE FOR CERTAIN CANCERS

- The IIT Madras has released a genome database covering paediatric leukaemia, colorectal and pancreatic cancers.
- The Bharat Cancer Genome Atlas (BCGA) has been made publicly accessible to researchers and clinicians in India and abroad.
- IIT Madras had launched the Indian Cancer Genome Programme in 2020, completing whole genome sequencing of patient samples collected nationwide, with principal support from Hyundai Motor India's Hyundai Hope for Cancer initiative, including Rs. 56 crore in funding.
- The research is led by IIT Madras Centre of Excellence on Cancer Genomics and Molecular Therapeutics, involved collaborations with Karkinos Healthcare, Mumbai, and multiple Chennai hospitals.
- Alongside BCGA, the institute also launched the Bharat Cancer Genome Grid (BCG2), a clinician-centric platform to integrate genomic data into routine oncology practice.

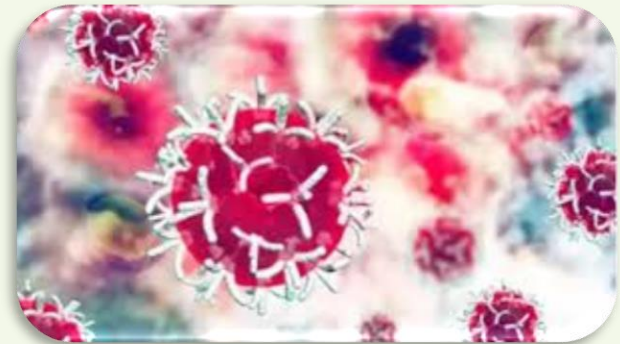
SUPPORTED BY DST, GOI

The initiative, supported by the Department of Science and Technology and the National Center for Precision Medicine in Cancer, also includes screening camps, technician training, and mobile healthcare outreach.



ONE IN NINE PEOPLE LIKELY TO DEVELOP CANCER IN THEIR LIFETIME

- As per ICMR, one in every nine people in India is likely to develop cancer in their lifetime. Currently, 14,16,427 persons are living with cancer.
- According to the ICMR, there has been a 12.8% increase in cancer incidence every year since 2022.



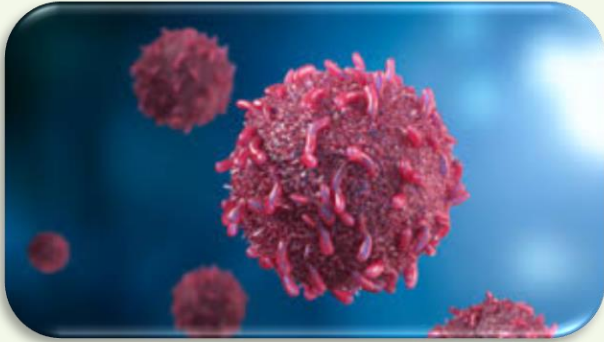
INDIA IS UNDER-REPRESENTED IN GLOBAL GENOME STUDIES



- Despite a high cancer incidence, India is under-represented in global genome studies, as per researchers. Efforts were not made to capture and catalogue the specific genetic variants of Indian cancers for want of genome architecture of cancers prevalent in the country. As a result, there have been little efforts to develop diagnostic kits and/or drugs.
- IIT Madras National Cancer Tissue Biobank, which launched the Cancer Genome Program in 2020, is to fill the gap in genome landscape for various cancers.
- The Tissue Biobank, (a joint initiative of the DST of the Union Government), is a non-profit community-backed facility in the IIT Madras Biotechnology Department. The Biobank collects cancer tissue samples with consent from cancer patients along with information about previous medical history.
- The samples and information are provided to researchers to facilitate cancer research and thus improve cancer diagnosis and treatment.



- As per IIT Director, the aim of making the database public is true to the institute's "IITM for all" commitment to society.
- Researchers can delve into the data and gain insights into the causes behind cancers and help devise prevention and early intervention initiatives.
- The Atlas also provides details of contemporary cancer populations to classify variants involved in early diagnostics, disease progression and treatment outcomes.
- The database would be an invaluable resource to identify cancer-specific biomarkers in India, enabling early detection of cancers. It will also be very useful to identify novel drug targets for developing better treatment strategies specific to the Indian population.
- The Institute's Centre of Excellence on Cancer Genomics and Molecular Therapeutics is funded under the Union Government's "Institution of Eminence" Initiative.



- The Atlas provides knowledge on the genetic basis of cancer progression and evolution and is expected to help biomedical research and healthcare system in the country.
- It also signals a shift to personalized medicine, whereby an individual's genetic and molecular information is used in the clinical decision-making process.
- Ultimately, this leads to the development of affordable cancer care solutions.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299